YEAR 8 MATHS · CHAPTER 1 · MIND MAP

Computation with integers

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1 Adding and subtracting

Line up the place values, then carry or borrow. Rounding first can make it easier in your head.

$347 + 586 = 933 \quad 502 - 178 = 324$

In your head: $198 + 57 = 200 + 57 - 2 = 255$

Estimate first: $347 + 586 \approx 350 + 590 = 940$

2 Multiplying and dividing

Split the number into parts you can multiply, then add the parts.

$34 \times 26 = 34 \times 20 + 34 \times 6 = 680 + 204 = 884$

$756 \div 7 = 108$

3 Number properties

Order doesn't matter for + and ×: $a + b = b + a$. Brackets can be expanded: $a(b + c) = ab + ac$.

$7 \times 23 = 7 \times 20 + 7 \times 3 = 161$

12 and 18: HCF is 6, LCM is 36

4 Divisibility and prime factorisation

Divisible by 2: even. By 3: digits add to a multiple of 3. By 5: ends in 0 or 5. By 9: digits add to a multiple of 9.

Keep dividing by primes: $60 = 2^2 \times 3 \times 5$ **and** $84 = 2^2 \times 3 \times 7$

1 is not a prime number. 2 is the only even prime.

5 Negative integers

Numbers get smaller as you move left on the number line.

$-5 < -2 \quad -1 > -8$

A temperature of −3°C is colder than 2°C

In order: $-8, -3, 0, 2$

A rise of 5° from $-3°$C gives $2°$C

6 Adding and subtracting negatives

Adding a negative means subtract. Subtracting a negative means add.

$-3 + 7 = 4 \quad 5 + (-8) = -3$

$-2 - (-6) = -2 + 6 = 4$

$-4 - 3 = -7$

7 Multiplying and dividing negatives

Same signs give a positive. Different signs give a negative.

$-4 \times 6 = -24 \quad -36 \div (-9) = 4$

$-2 \times (-3) \times (-1) = -6$: an odd number of negatives gives a negative

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$

8 Order of operations and substitution

Brackets, then powers, then × and ÷ from left to right, then + and − from left to right.

$2 + 3 \times (-4) = 2 - 12 = -10$

If $a = -2$ and $b = 3$: $a^2 - b = 4 - 3 = 1$

Watch out: put negative numbers in brackets when you substitute.